CodonRescore: Annotation-Free Codon-Model Rescoring of Ab Initio Gene Calls

A benchmark on organellar and small prokaryotic genomes

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Bioinformatics Research Work — Internal Assessment

Problem and motivation

Gene prediction = first step of annotation; pipelines such as Bakta delegate CDS finding to Prodigal/Pyrodigal.

Self-trained finders can degrade on short, AT-rich, RNA-gene-rich genomes (e.g. chloroplasts).

Deep ORF classifiers (Balrog, ProtiGeno) need large pre-training; labs need a cheap, genome-specific check.

Question: can a codon model trained only on the genome itself clean up a caller's output — and when does it fail?

Literature survey (recent work, 2021–2024)

Work	Approach	Limitation for us
Pyrodigal (2022)	Prodigal port; self-training / meta models	Needs ≥20 kb to train
Balrog (2021)	Universal protein model (TCN)	Heavy pre-training
Dimonaco et al. (2022)	Benchmark of gene finders	Organism-dependent accuracy
Bakta (2021)	Annotation pipeline on Prodigal CDS	Inherits CDS errors
ProtiGeno (2023)	ESM-1b embeddings for short ORFs	Lower precision; large model
BRAKER3 (2024)	Evidence integration (eukaryotes)	Needs RNA-seq/protein data
DNABERT-2 (2024)	DNA foundation model	Not tested for gene calling

Research gap

G1 Self-training on small organellar genomes is not examined in the works reviewed.

G2 Cheap, genome-specific post-hoc rescoring (no pre-training, no annotation) is little explored.

G3 Error anatomy of false positives is rarely reported — where should rescoring intervene?

Datasets (real NCBI records)

Record	Length	GC	CDS (all / eval.)	Role
A. thaliana chloroplast NC_000932	154,478	36.3%	85 / 70	Truth + training
Y. pestis plasmid pPCP1 NC_005816	9,609	45.3%	10 / 10	Truth
Phage phiX174 NC_001422	5,386	44.8%	11 / 8	Truth
C. diphtheriae NCTC11397	2,463,666	53.5%	no annotation	Scalability only

Scoring: stop-codon + strand match; precision / recall / F1; bootstrap 95% CI. Annotation is used for scoring only — never for training or tuning.

Methodology: CodonRescore

1 Enumerate ORFs (≥90 nt, both strands); seeds = longest, low-overlap ORFs (≥450 nt).

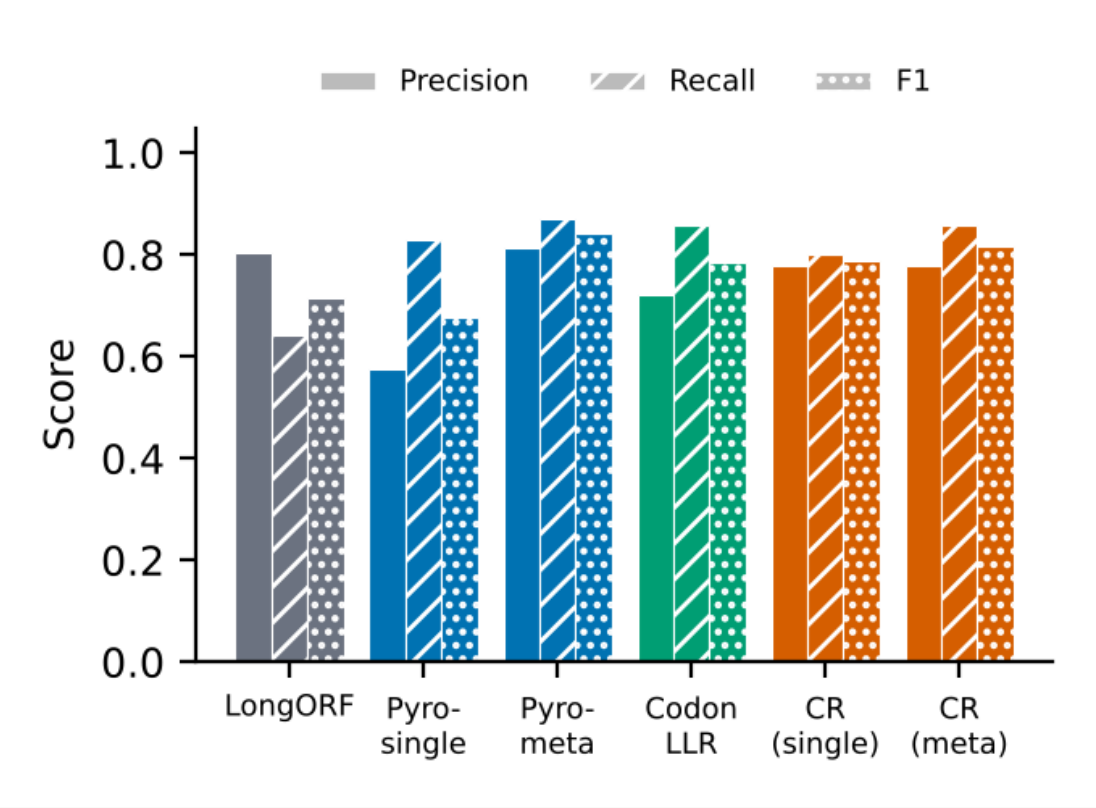
2 3-periodic order-2 Markov coding model vs. genome background → per-codon log-likelihood ratio S.

3 Threshold τ: 2-fold cross-fitted Youden cut-off, held-out seeds vs. shadow-frame ORFs.

4 Re-score caller output: prune short calls with S<τ (<300 nt); rescue uncalled ORFs with S≥τ (≥120 nt).

Applied to Pyrodigal single mode (self-trained) and meta mode (pre-trained).

Results — chloroplast genome (70 evaluable genes)



Method	Prec.	Rec.	F1
Pyrodigal single	0.574	0.829	0.678
Pyrodigal meta	0.813	0.871	0.841
CodonRescore (single)	0.778	0.800	0.789
CodonRescore (meta)	0.779	0.857	0.816

Single mode: 43 false positives → 16 after rescoring.

F1 95% CIs overlap (0.59–0.75 vs 0.72–0.85).

Also: pPCP1 and phiX174 — Pyrodigal-meta best; CodonRescore not applicable on phiX174 (too few seeds).

Critical analysis

Of 43 single-mode false positives: 38 < 300 nt, 24 overlap tRNA/rRNA genes, none shadow a true protein gene.

All methods find every gene ≥600 nt; differences are in the 25 genes <300 nt.

Rescue step hurts: 0 of 10 rescued ORFs true in meta mode; order-2 model is too weak for rescue.

Threshold is brittle: lowering τ by 0.1 drops F1 below 0.44; conservative is safer.

Only 70 truth genes → differences mostly not significant; no hyper-parameter tuning was done.

Proposed research extension

Validate on many annotated organelle genomes + bacterial chromosomes for statistical power.

Richer scorer: hexamer / higher-order Markov, start-site and RBS features.

Protein-level evidence: protein / DNA language-model embeddings (ESM-1b, DNABERT-2) as rescoring features.

Calibrated classifier instead of a single Youden cut-off; prune-only variant.

Head-to-head with Balrog and ProtiGeno; extend to spliced genes and eukaryotic pipelines (BRAKER3 ideas).

Conclusion

Benchmarked Pyrodigal on real organellar / small genomes against RefSeq annotation.

CodonRescore (annotation-free) cuts false positives of self-trained Pyrodigal 43 → 16 (F1 0.678 → 0.789).

No improvement over pre-trained meta mode (0.816 vs 0.841) — and evidence is limited by 70 genes.

Practical advice: prefer pre-trained/conservative settings on organelle genomes; distrust short, low-scoring calls near RNA genes.

Manuscripts prepared in IEEE and Springer LNCS formats; code and results provided.